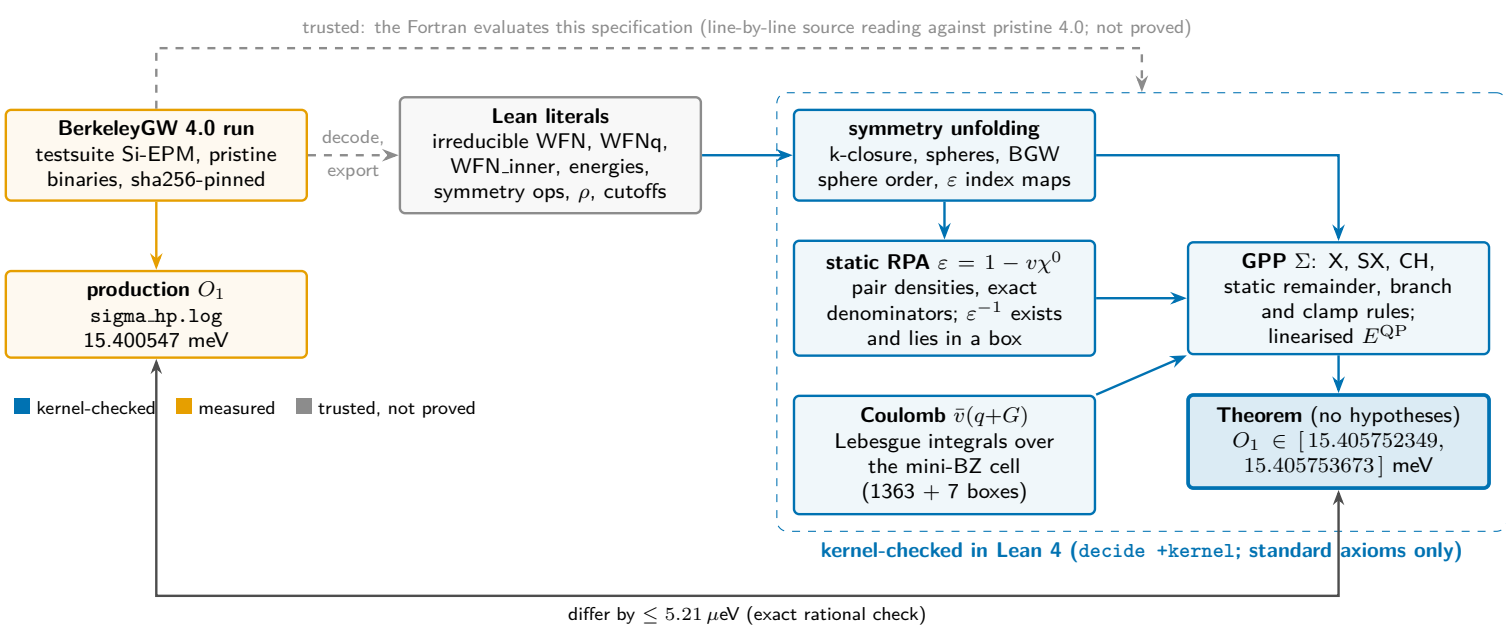

trusted: the Fortran evaluates this specification (line-by-line source reading against pristine 4.0; not proved)
BerkeleyGW 4.0 run
testsuite Si-EPM, pristine
binaries, sha256-pinned
decode,
export
Lean literals
irreducible WFN, WFNq,
WFN_inner, energies,
symmetry ops, ρ, cutoffs
symmetry unfolding
k-closure, spheres, BGW
sphere order, ε index maps
static RPA ε = 1 − vχ0
pair densities, exact
denominators; ε−1 exists
and lies in a box
GPP Σ: X, SX, CH,
static remainder, branch
and clamp rules;
linearised EQP
Coulomb v̄(q+G)
Lebesgue integrals over
the mini-BZ cell
(1363 + 7 boxes)
Theorem (no hypotheses)
O1 ∈ [15.405752349,
15.405753673] meV
kernel-checked in Lean 4 (decide +kernel; standard axioms only)
production O1
sigma_hp.log
15.400547 meV
kernel-checked
measured
trusted, not proved
differ by ≤ 5.21 μeV (exact rational check)